Lecture 5B - When Complete Randomization Fails

Attrition after Random Assignment

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Outline

1. Outcome missingness after random assignment
2. Bounding approach
3. Bounding approach in attrition
4. Conclusing remarks

When complete randomization fails

Randomization alone does not guarantee that we observe a trustworthy outcome for every unit.

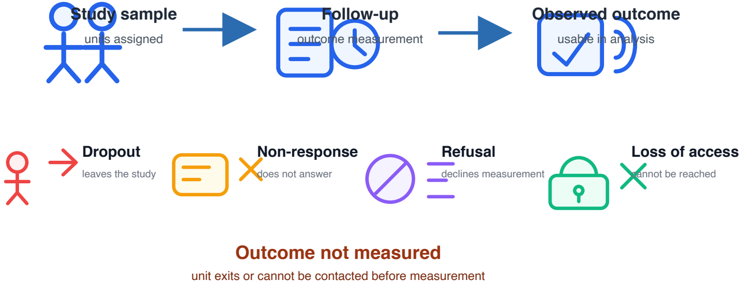
Core problem

After treatment assignment, some outcomes become **unobserved** or **unusable**. We only observe outcomes for units that remain in the analysis sample.

The key question: how do we analyze the data?

Two sources of post-randomization outcome missingness

Natural attrition: Outcomes are not measured because units leave the study or cannot be reached.



Natural attrition: an example

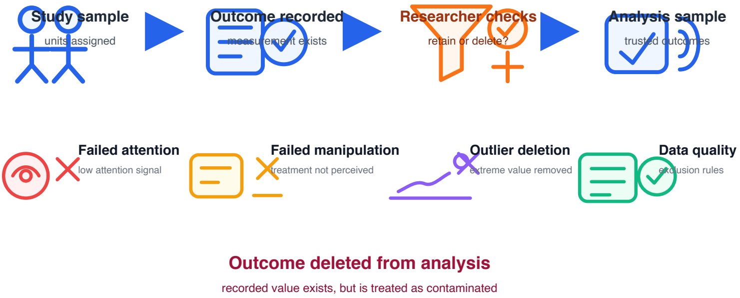
The introduction of e-commerce to countryside in China[1].

Attrition: **local governments denied access to some villages.**

[1] Please check Couture et al. (2021) for more details.

Two sources of post-randomization outcome missingness

Artificial attrition: Outcomes are measured, but deleted because they are judged unreliable.



Both create a selected analysis sample.

Why put them in the same category?

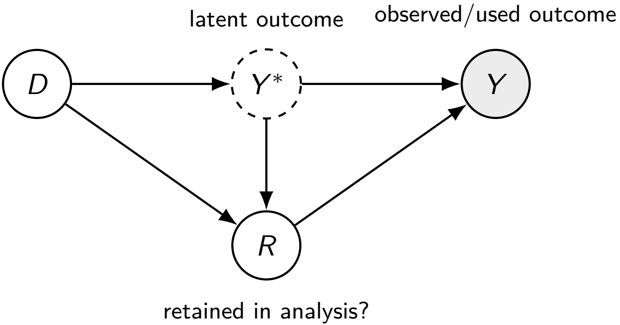
The difference is substantive:

- Natural attrition: the outcome is unavailable.
- Artificial attrition: the measured outcome is considered contaminated.

Common logic

For the treatment effect, both cases mean that the trustworthy latent outcome Y_i^* is only partially observed.

A DAG for natural and artificial attrition



- D: randomized treatment.
- Y^*: the outcome we want for causal inference.
- $R = 1$: unit is removed from the analysis sample.
- Y: outcome actually used in estimation.

The identification problem

If we estimate with the treatment effect with the retained samples, what do we get?

$$\underbrace{E(Y_i \mid D_i = 1, R_i = 0) - E(Y_i \mid D_i = 0, R_i = 0)}_{\text{difference among retained observations}}$$

vs.

$$\underbrace{E(Y_i \mid D_i = 1) - E(Y_i \mid D_i = 0)}_{\text{difference in the full assigned sample}}$$

Failed identfication

With no assumption $\Leftrightarrow$ Intent-to-treat (hard to interpret) or point identification failed (if Y^* depends on unobservables).

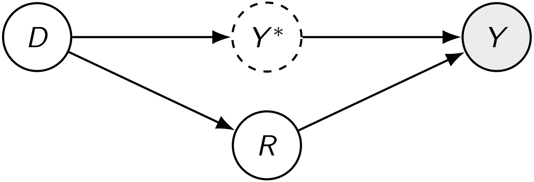
$$
\begin{tikzcd}
D \arrow[r] \arrow[dr] & Y^* \arrow[r] \arrow[d] & Y \\
& R \arrow[ur] &
\end{tikzcd}
$$

Fact (Intention-to-treat)

Without any assumption, we can at most identify the intention-to-treat effects with $P(Y \mid D)$.

Traditional approaches: Random attrition

Make a strong assumption of random attrition or attrition is at random.



Fact (Random Attrition)

Under random attrition, we can identify the average treatment effect with $P(Y^* \mid D) = P(Y^* \mid D, R)$.

Traditional approaches: Random attrition

Technical proof:

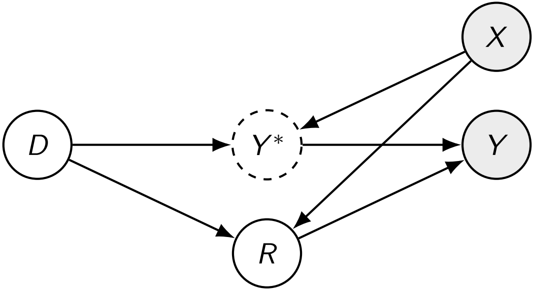
$$\begin{aligned} P(Y^* \mid do(D)) &= P(Y^* \mid D, R=0)\, P(R=0) + P(Y^* \mid D, R=1)\, P(R=1) \\ &\overset{Y^* \perp R \mid D}{=} P(Y^* \mid D)\, P(R=0) + P(Y^* \mid D)\, P(R=1) \\ &= P(Y^* \mid D) \end{aligned}$$

Because the attrition is at random, we can use observed outcome Y to calculate the conditional distribution:

$$P(Y^* \mid D) = P(Y^* \mid D, R=0) = P(Y \mid D)$$

Traditional approaches: imputation

Make an assumption that we observe **non-descendant** variables X that **d-separate** Y^* **and** R.



Fact (Imputation Method)

Under the assumption of the imputation method, we can identify the average treatment effect with
$P(Y^* \mid D, X) = P(Y^* \mid D, X, R)$.

Traditional approaches: imputation

Technical proof:

$$
\begin{aligned}
P\left(Y^{*} \mid do\left(D\right), X\right) &= P\left(Y^{*} \mid D, X, R=0\right) P\left(R=0\right) \\
&\quad + P\left(Y^{*} \mid D, X, R=1\right) P\left(R=1\right) \\
&\overset{Y^{*} \perp R \mid X}{=} P\left(Y^{*} \mid D, X\right) P\left(R=0\right) + P\left(Y^{*} \mid D, X\right) P\left(R=1\right) \\
&= P\left(Y^{*} \mid D, X\right)
\end{aligned}
$$

Because the attrition is at random conditional on X, we can use observed outcome Y to calculate the conditional distribution:

$$
P\left(Y^{*} \mid D, X\right) = P\left(Y^{*} \mid D, X, R=0\right) = P\left(Y \mid D, X\right)
$$

We can then integrate over X to have the ATE:

$$
P\left(Y \mid D, X\right) = \sum_{x \in X} P\left(Y \mid D, X=x\right)
$$

Issues of the traditional approaches

The law of decreasing credibility

The credibility of inference decreases with the strength of the assumptions maintained.

Assumptions are strong and cannot be justified.
Credibility is sacrificed for the "power" of conclusions.

- The random attrition is hardly a credible assumption.
- Difficult to find a set of X that $\underline{d\text{-seperate } Y^* \text{ and } R}$.

Could we adopt a better strategy?

1. Outcome missingness after random assignment
2. **Bounding approach**
3. Bounding approach in attrition
4. Conclusing remarks

Relaxing unrealistic assumptions

Many assumptions are non-refutable and unrealistic.

- Unconfoundedness
- Random attrition
- ...

Forego the point identification, make weaker assumptions and increase credibility of analyses.

Strong assumptions identify a point: •

Weak assumptions identify a set: [– – – –]

Some types of bounds: no-data bounds

Assume the potential outcomes Y^0 and Y^1 are bounded $[0,1]$.

$$-1 \leq Y_i^1 - Y_i^0 \leq 1 \Rightarrow E\left(Y_i^1 - Y_i^0\right) \in [-1,1]$$

In more general terms, if Y^0 and $Y^1 \in [a,b]$

$$E\left(Y_i^1 - Y_i^0\right) \in [a-b, b-a]$$

The length of the bound is $2\,(b-a)$.

Some types of bounds: no-assumption bounds

Assume we collect some data with $\{Y_i, D_i\}$, but random assignment is not ensured[2].
Can we do better with data?

$$E\left(Y_i^1 - Y_i^0\right) = \underbrace{\pi E\left(Y_i^1 \mid D_i = 1\right)}_{\text{Observed}} + (1-\pi)\underbrace{E\left(Y_i^1 \mid D_i = 0\right)}_{\text{Unobserved}}$$
$$- \pi \underbrace{E\left(Y_i^0 \mid D_i = 1\right)}_{\text{Unobserved}} - \underbrace{(1-\pi) E\left(Y_i^0 \mid D_i = 0\right)}_{\text{Observed}}$$

Under the bounds on potential outcomes $[a, b]$:

$$\tau^{obs} + (1-\pi) a - \pi b \leq E\left(Y_i^1 - Y_i^0\right) \leq \tau^{obs} + (1-\pi) b - \pi a$$

The length of the bound is $(b-a)$, half of the no-data bounds.

[2] Let $\pi = P(D_i = 1)$ in the data collected.

Some types of bounds: monotone treatment response

Assume treatment always benefits people, or $Y_i^1 \geq Y_i^0, \forall i$
This means ITE is nonnegative, so the lower bound of ITE is 0.
Under this assumption, the lower bound of ATE is also 0:

$$E\left(Y_i^1 - Y_i^0\right) \geq 0$$

Using this assumption, we may further narrow the no-data and no-assumptions bounds.

Fact

Under monotone treatment response assumption, the no-data bound becomes $E\left(Y_i^1 - Y_i^0\right) \in [0, b - a]$.

Some types of bounds: monotone treatment selection

Assume the treatment group potential outcomes are better than control groups, with:

$$E\left(Y_i^1 \mid D_i = 1\right) \geq E\left(Y_i^1 \mid D_i = 0\right)$$
$$E\left(Y_i^0 \mid D_i = 1\right) \geq E\left(Y_i^0 \mid D_i = 0\right)$$

Under this assumption, the ATE is bounded from above,

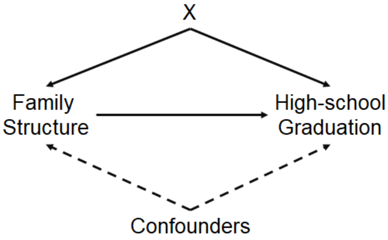
$$E\left(Y_i^1 - Y_i^0\right) \leq E\left(Y_i^1 \mid D_i = 1\right) - E\left(Y_i^0 \mid D_i = 0\right)$$

We may use it to further narrow the no-assumption bounds.

Bounding approach in practice

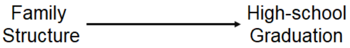
Manksi et al. (1992) studied how family structure during adolescence influences the high school graduation rate.

Data from National Longitudinal Survey of Youth: D (if living with both parents), Y (if graduate by 20), X (a set of control variables).

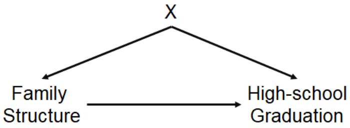


Assumption 1 & 2: No confounders / Selection observables

Variable	**Exogenous**	**Selection on X**
Constant	0.47 (5.81)	0.51 (11.61)
Race and ethnicity		
Parental education		
Residence		
Nonintact family at 14	–0.36 (6.75)	-0.38 (2.27)



(a) No confounders



(b) Selection on X

Family structure: no assumption bounds

	Intact Family	Non-Intact Family
Graduated	**1**	**2**
Not Graduated	**3**	**4**

- **Lower bound**: Flipping the outcome of in Cell 1 and 4.
- **Upper bound**: Flipping the outcome of in Cell 2 and 3.

Estimation results: $[-0.74, +0.20]$.

Family structure: monotone treatment selection

For any value of unobserved confounders, any youth when put in an intact family, are at least as likely to graduate as when not.

$$E\left(Y_i^1 \mid D_i = 1\right) \geq E\left(Y_i^1 \mid D_i = 0\right)$$
$$E\left(Y_i^0 \mid D_i = 1\right) \geq E\left(Y_i^0 \mid D_i = 0\right)$$

Estimation results: $[-0.31, -0.02]$.

Family sturcture: comparing difference approaches

Approach	Estimation
Exogenous	–0.36
Selection on Observables	-0.38
Monotone Treatment Selection	[–0.31, –0.02]
No Assumptions Bounds	[–0.74, +0.20]

Law of Decreasing Credibility

Stronger conclusiosn require stronger assumptions that have a lower less of credibility.

A running example

- We have a data from a randomized experiment.
- The outcome Y_i measured with a scale from 1 to 10 (i.e., $Y_i \in [1, 10]$).
- The experiment has an attrition problem, with NA denoting missing values.

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

$Y_i \in [1, 10]$

A running example

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

$Y_i \in [1, 10]$

- Without additional assumptions, the intention-to-treat is: $\text{ITT}_Y = \frac{10+8}{3} - \frac{3+6}{3} = 3$.
- Under random attrition, the ATE is: $\text{ATE}_{\text{random}} = \frac{10+8}{2} - \frac{3+6}{2} = 4.5$.

Bounding approach: Manski bounds

Manski bounds have no additional assumptions. See Horowitz and Manski (2000).

- Upper bound: substituting NA's in **treatment with max values and control with min values**.
- Lower bound: substituting NA's in **treatment with min values and treatment with max values**.

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

$Y_i \in [0, 10]$

Bounding approach: Manski bound

Obs	D_i	Y_i
1	1	10
2	1	8
3	1	NA
4	0	3
5	0	NA
6	0	6

- Manski bound: $\left[\frac{10+8+1}{3} - \frac{3+10+6}{3}, \frac{10+8+10}{3} - \frac{3+1+6}{3}\right] = [0, 6]$
- $\text{ITT}_Y = 3$ and $\text{ATE}_{\text{random}} = 4.5$

In practice, use *max* and *min* values of observed outcomes.
Uninformative but wide, as a benchmark for further analysis.
Can we narrow the bound?

Lee bounds Lee (2009)

Assumptions: **monotonicity of treatment selection**.

- Treatment status either increases / decreases attrition for all people.
- The effect of treatment on attrition has the same sign ($+$ or $-$).

How to use the condition?

- Trimming samples such that share of observed individuals is equal in the treatment and control.
- Trimming from above or below to obtain lower or upper bound.

Lee bounds: key idea

- Suppose we have a treatment group with 40% of attrition and a control group with 50%.

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- To get the same level of "missing," we need to trim x% of people from the treatment.

Lee bounds: key idea

- Suppose we have a treatment group with 40% of attrition and a control group with 50%.
- The treatment decreases the attrition likelihood for 10%.
- To get the same level of "missing," we need to trim x% of people from the treatment.
- To calculate x%, we do:

$$\frac{40\%}{1 - x\%} = 50\% \Rightarrow x\% = \frac{50\% - 40\%}{50\%} = 20\%$$

Lee bounds: key idea

Question: **which 20% of the treated units to trim?**

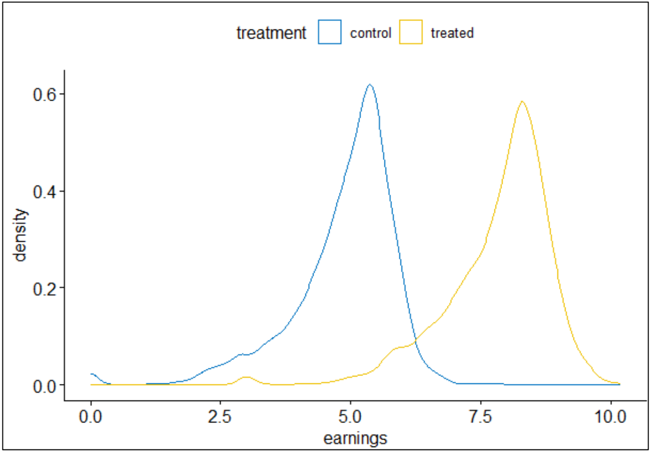
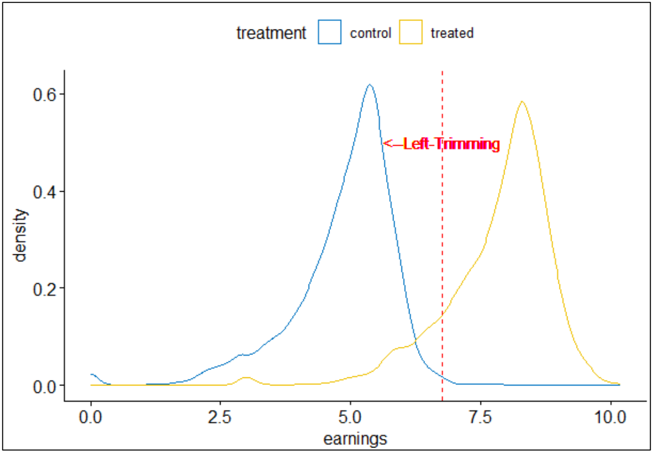


Figure 2: Density Plots of Outcomes for the Treated and Control Group

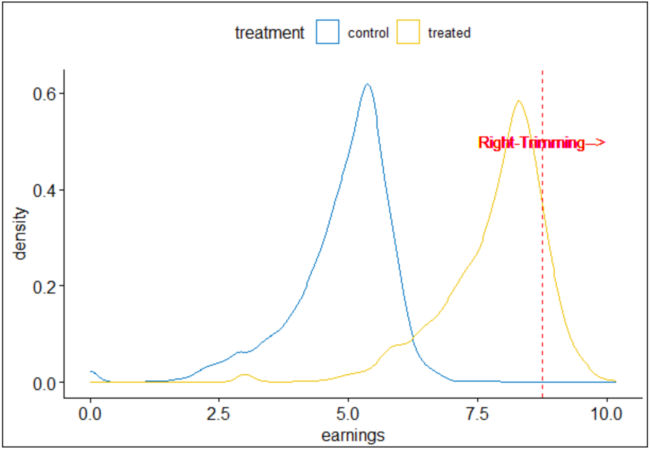
Lee bounds: key idea

To trim the people in the 20% lower quantile to get the upper bound:



Lee bounds: key idea

To trim the people in the 20% upper quantile to get the lower bound:



Lee bounds: procedure

Y the outcomes, D the treatment, and W the non-attrition.

- Step 1: Get the share of individuals with observed Y in the treatment and control, as q_1 and q_0.
- Step 2: Obtain the attrition difference:

$$q = \begin{cases} \frac{q_1 - q_0}{q_1} & \text{if } q_1 > q_0 \\ \frac{q_0 - q_1}{q_0} & \text{if } q_0 > q_1 \end{cases}$$

- Step 3: Find the qth and $(1-q)$th quantile of observed outcomes in the treatment group as Y^1_q and Y^1_{1-q}.

Lee bounds: procedure

Y the outcomes, D the treatment, and W the non-attrition.

- Step 4: Upper and lower bound:

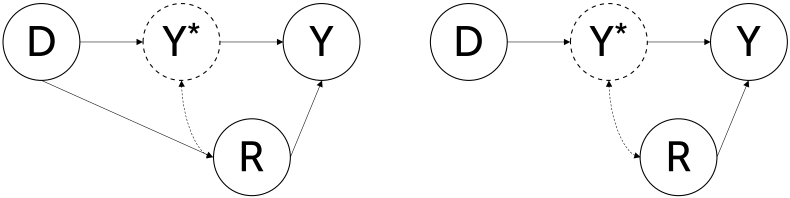
$$\overline{\tau} = \underbrace{\frac{\sum 1\left(Y_i \geq Y^1_q\right) D_i W_i Y_i}{\sum 1\left(Y_i \geq Y^1_q\right) D_i W_i}}_{\text{Non-attritors in treatment with higher outcome}} - \underbrace{\frac{\sum (1 - D_i) W_i Y_i}{\sum (1 - D_i) W_i}}_{\text{Non-attriters in control}}$$

$$\underline{\tau} = \underbrace{\frac{\sum 1\left(Y_i \leq Y^1_{1-q}\right) D_i W_i Y_i}{\sum 1\left(Y_i \leq Y^1_{1-q}\right) D_i W_i}}_{\text{Non-attritors in treatment with lower outcome}} - \underbrace{\frac{\sum (1 - D_i) W_i Y_i}{\sum (1 - D_i) W_i}}_{\text{Non-attriters in control}}$$

Lee bounds: summary

Treatment influences attrition in one direction:

- By comparing the treatment and control, we may "partial out" the effect of treatment on attrition.
- ATE on the non-attritors!



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Why bounding approaches are needed?

Current common practice: “delete and run”

In both natural and artificial attrition, researchers often remove problematic observations and estimate the treatment effect on the remaining sample.

This practice is convenient, but it changes the estimand due to sample selection

Main concern

A complete-case estimate is, at best, an effect for the retained sample. In many cases, it is a biased estimate of the finite population ATE.

From deletion to sensitivity

The current norm is often:

delete observations $\longrightarrow$ run the main model.

But for post-treatment deletion, especially artificial attrition, this is no longer sufficient.

Future practice should make the consequences of deletion visible:

- report the main complete-case estimate;
- estimate bounds under weak assumptions;
- show whether conclusions are consistent across assumptions.

References I

- Couture, Victor, Benjamin Faber, Yizhen Gu, and Lizhi Liu. 2021. "Connecting the Countryside via E-Commerce: Evidence from China." American Economic Review: Insights, 3 (1): 35-50.
- Horowitz, J. L., & Manski, C. F. (2000). Nonparametric analysis of randomized experiments with missing covariate and outcome data. Journal of the American Statistical Association, 95(449), 77-84.
- Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. The Review of Economic Studies, 76(3), 1071-1102.

References II

Manski, C. F., Sandefur, G. D., McLanahan, S., & Powers, D. (1992). Alternative estimates of the effect of family structure during adolescence on high school graduation. Journal of the American Statistical Association, 87(417), 25-37.